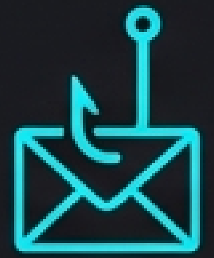
CRYPTOGRAPHY

THE SCIENCE OF SECRETS

Grade 7 Computer Science // Top Secret Clearance

THE MISSION: PROTECTING YOUR DATA

The Interceptor
A third party trying to steal private messages.



Phishing
Fake emails tricking you into sharing passwords.

Man-in-the-Middle
Hackers secretly tracking data packets.

WHAT IS CRYPTOGRAPHY?

Definition: The science of encoding and decoding information to safeguard its integrity.

Key Terms:

- **Encoding:** Storing information inside something else (like a secret handshake or code).
- **Decoding:** The process of understanding the hidden information.
- **Cryptography:** The digital disguise for our data.

THE FOUR PILLARS OF SAFETY

CONFIDENTIALITY
(Only the right people see it)

DATA PRIVACY
(Safeguarding sensitive info)

AUTHENTICATION
(Are you who you say you are?)

SECURITY
(Protection against threats)

ANCIENT SECRETS: THE CAESAR CIPHER

History: Used by Julius Caesar to send secret orders to his armies.

The Logic: Shift every letter **3 steps** to the **right**.

Example: **A** becomes **D**, **B** becomes **E**.

AGENT CHALLENGE: CRACK THE CODE

CLASSIFIED DOSSIER – CARD 1

CODE:

CVVCEM CV FCYP

HINT:

→ Shift letters -2 spaces

ANSWER

CONFIDENTIAL

ANSWER:
ATTACK AT DAWN

CLASSIFIED DOSSIER – CARD 2

CODE:

TAERGEHTREDNAXELA

HINT:

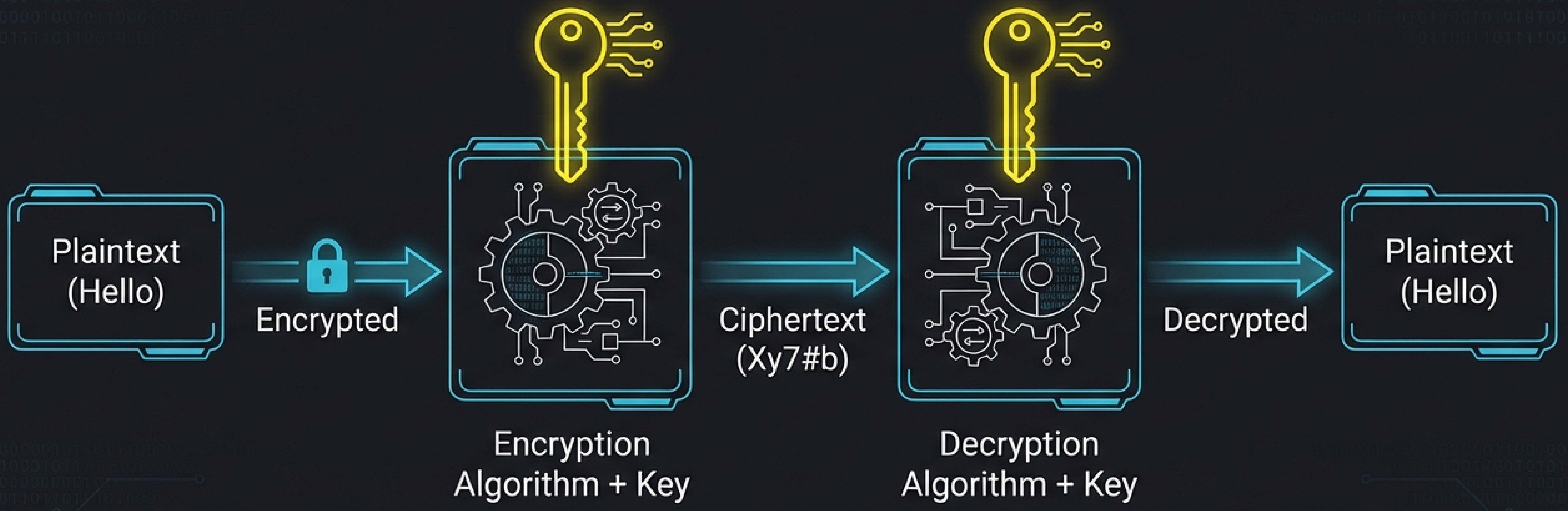
→ Reverse the letters

ANSWER

CONFIDENTIAL

ANSWER:
ALEXANDER THE GREAT

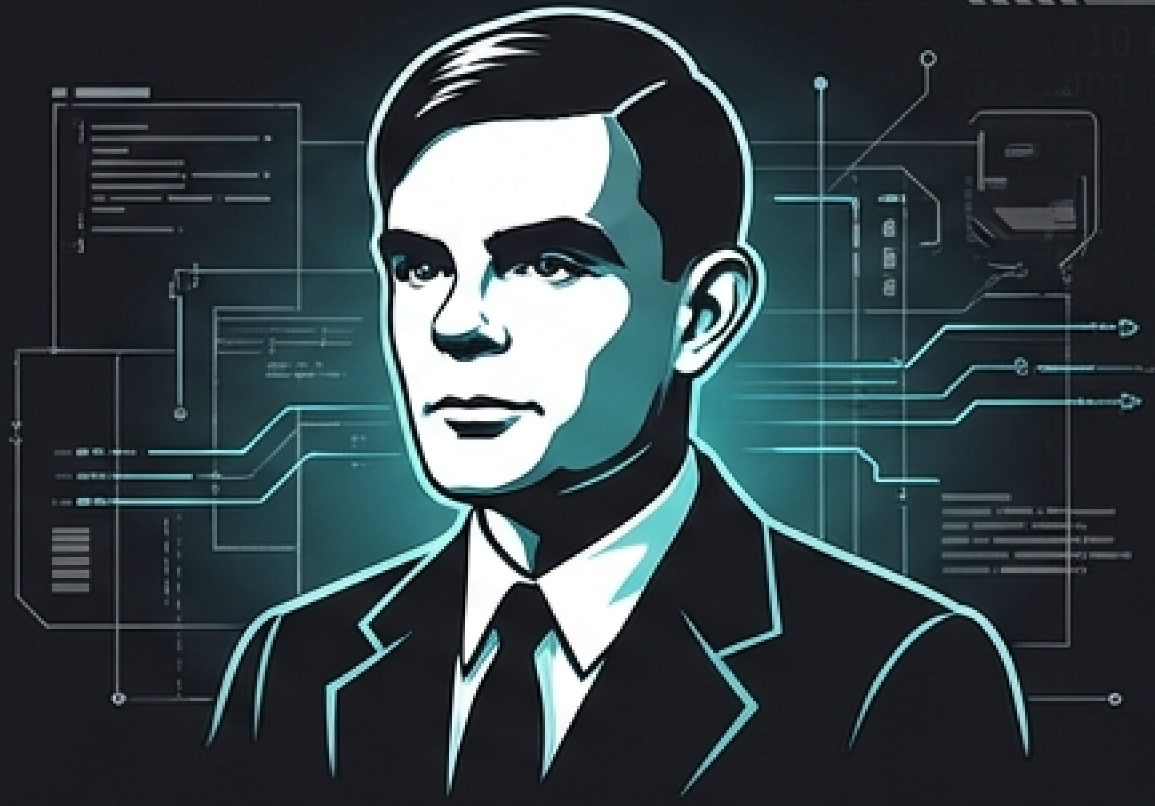
ANATOMY OF A CRYPTOSYSTEM



The Cryptosystem Model

THE CODE THAT CHANGED HISTORY

The **Enigma Machine:** Used in WWII for complex encryption.



Alan Turing: The Father of Computer Science.

Impact: Invented the 'Bombe' machine to break Enigma, paving the way for modern computers.

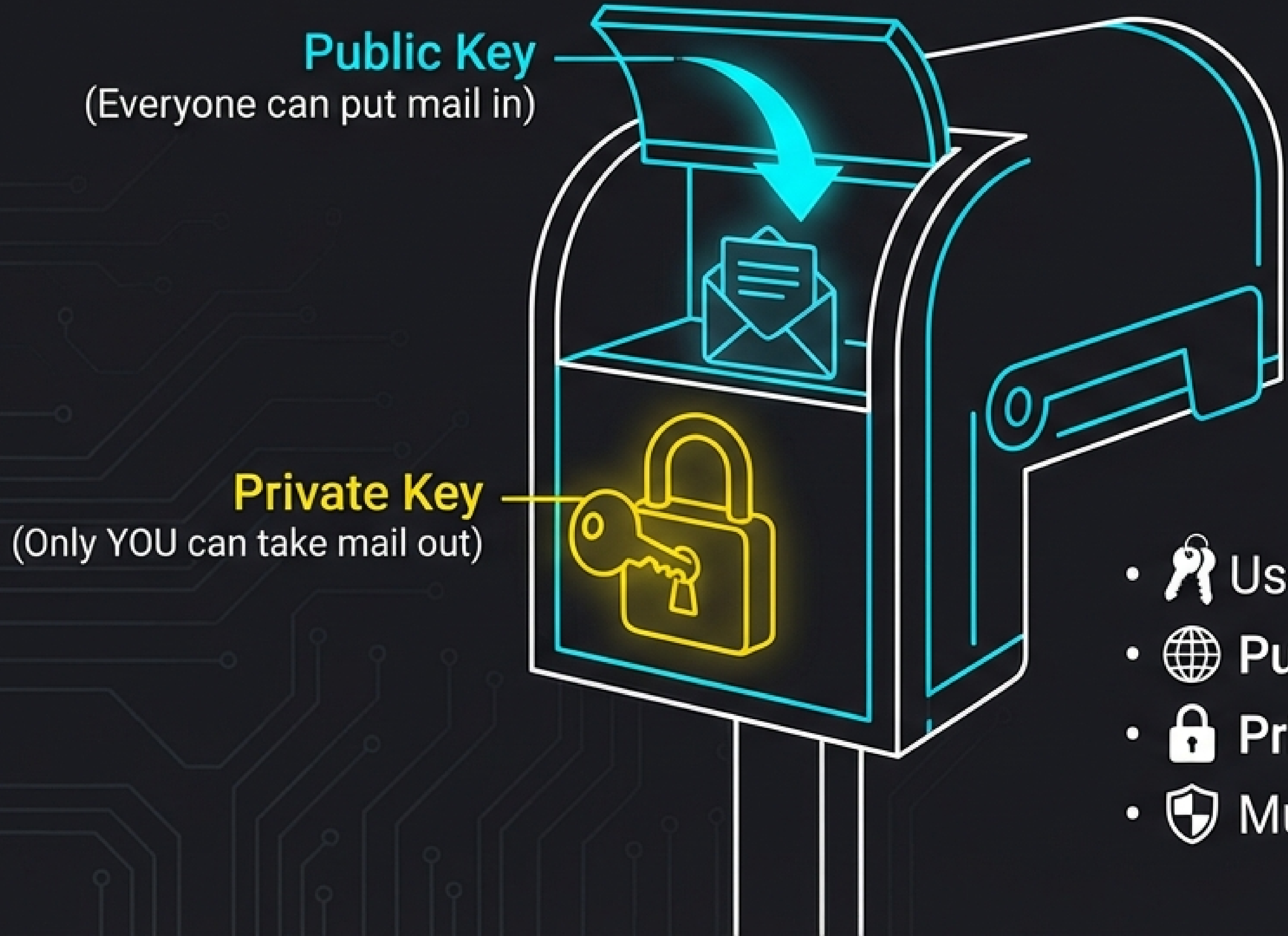
TYPE 1: SYMMETRIC KEY ENCRYPTION

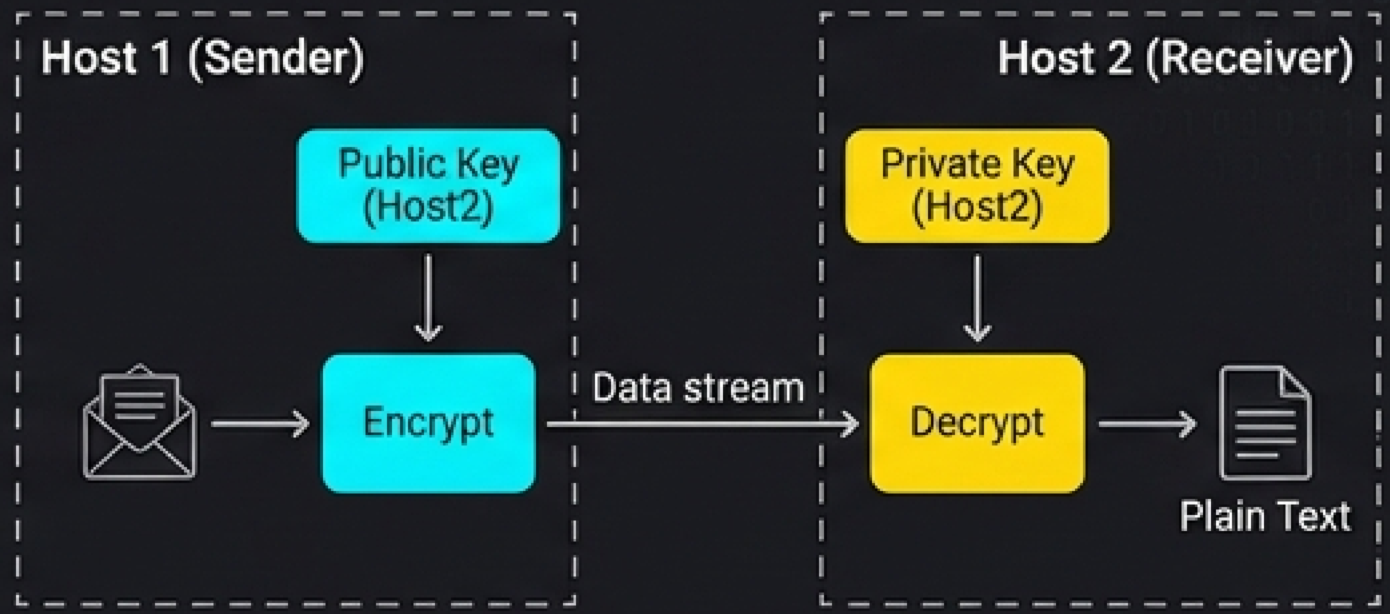
The 'Shared Secret' Method

- Same Key used to Lock (Encrypt) and Unlock (Decrypt).
- Fast and simple.
- Risk: If you share the key, it might be stolen!

TYPE 2: ASYMMETRIC KEY ENCRYPTION

The 'Public & Private' Method





- Uses **TWO** different keys.
- **Public Key**: Available to everyone.
- **Private Key**: Kept secret by the owner.
- Much safer—you never share your private key.

BATTLE OF THE KEYS

SYMMETRIC

- 1 Key (Shared)
- Faster Speed
- Best for: Bulk Data

ASYMMETRIC

Public Private

- 2 Keys (Public + Private)
- Slower (Complex Math)
- Best for: Authentication & Digital Signatures

REAL WORLD: END-TO-END ENCRYPTION

1. Two keys generated when you open the app.

2. Message locks on your phone.
3. Travels securely through the server.

4. Only the receiver's private key can unlock it.

Not even the app company can read your messages.

PROTECTING YOUR MONEY

Online Banking

- Uses Encryption + OTP (One Time Password) or Biometrics.

Cryptocurrency

- Digital money created from code.
- Peer-to-Peer (No banks involved).
- 'Crypto' currency = Secured by Cryptography.

MISSION DEBRIEF: POP QUIZ

What is the unintelligible text called after encryption?

REVEAL ANSWER

CIPHERTEXT

Which encryption uses TWO different keys?

REVEAL ANSWER

ASYMMETRIC

Who is the father of Modern Computing?

REVEAL ANSWER

ALAN TURING

MISSION ACCOMPLISHED

Cryptography keeps the internet safe.
Keep your keys secret!

Presentation Complete.